IMPERIAL

An Intuitive Introduction to Stochastic Analysis

Generative Modelling with Stochastic Differential Equations

Jack Daly
November 2026

Contents

Summary. Motivate stochastic analysis as the study of random processes. Explain how to understand intuitively all the essential tools. Everything in the module is built on these tools.

Part I: Rapid introduction

01 Why stochastic analysis?
02 Brownian motion
03 Quadratic variation: second-order calculus
04 Filtrations, martingales, measurability
05 The Itô integral
06 Zero mean and the Itô isometry
07 Stochastic differential equations
08 Itô's formula
09 Further facts
10 Girsanov's theorem

Part II: First further theory

11 Markov semigroups, generators, invariant measures

Appendix

A Divergences, Fokker–Planck, proofs, Girsanov intuition

Part I

An intuitive introduction to stochastic analysis

Why stochastic analysis?

- **Goal**: model random processes $(X_t)_{t \geq 0}$ in continuous time.
- This needs a **probabilistic theory** of such processes: how to define them, integrate against them, and differentiate functions of them.

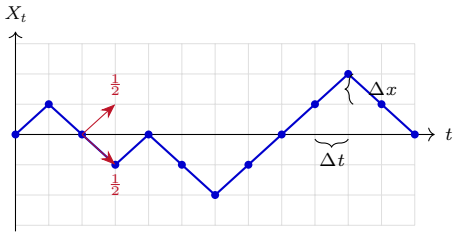
Strategy Start from **fundamental processes** that we can define precisely and build a theory around: **martingales** ("things we expect to stay the same") and **Brownian motion** (continuous-time "pure noise"). Then build **more complex random processes** from them, e.g. a stochastic differential equation

$$\mathrm{d}X_t = \underbrace{b_t(X_t)\,\mathrm{d}t}_{\text{drift}} + \underbrace{\sigma_t(X_t)\,\mathrm{d}B_t}_{\text{noise}}.$$

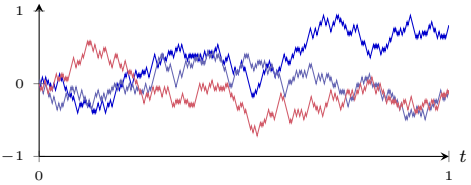
In this module: the Langevin diffusion and the noising / denoising processes of diffusion models are SDEs of this form.

Brownian motion: intuition

A symmetric random walk, refined



Every Δt, a fair coin moves the walk $\pm\Delta x$.



Three paths with $\Delta t = 1/500$, $\Delta x = \sqrt{\Delta t}$.

The $\mathrm{Var} \propto \mathrm{d}t$ condition After time t there are $t/\Delta t$ independent steps, so $\mathrm{Var}(X_t) = \frac{t}{\Delta t}(\Delta x)^2$. A non-trivial limit ($0 < \mathrm{Var}(X_t) < \infty$) as $\Delta t \to 0$ forces $(\Delta x)^2 \propto \Delta t$: **increments of size $\sqrt{\mathrm{d}t}$, variance $\mathrm{d}t$.** The limit is Brownian motion.

Brownian motion: definition

Definition (Brownian motion)

A process $B = (B_t)_{t \geq 0}$ in $\mathbb{R}^n$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a (standard, n-dimensional) **Brownian motion** if

1. $B_0 = 0$ almost surely;
2. **continuous paths**: $t \mapsto B_t(\omega)$ is continuous for $\mathbb{P}$-almost every ω;
3. **independent increments**: for all $0 \leq t_0 < t_1 < \cdots < t_m$, the increments $B_{t_1} - B_{t_0}, \ldots, B_{t_m} - B_{t_{m-1}}$ are independent;
4. **Gaussian increments**: $B_t - B_s \sim \mathcal{N}\big(0, (t-s) I_n\big)$ for all $0 \leq s < t$.

Intuition

- (iii)–(iv): the limiting $\mathrm{Var} \propto \mathrm{d}t$ condition beconmes $\mathrm{Var}(W_t - W_s) = (t - s)$
- The coordinates $B^1, \ldots, B^n$ are independent one-dimensional Brownian motions.
- Paths are continuous but have **infinite variation** on every interval: $\int H_t \, \mathrm{d}B_t$ cannot be a path-by-path Riemann–Stieltjes integral.

Square increments do not vanish

$\Rightarrow$ a second-order calculus

Proposition (quadratic variation of Brownian motion; proof in A3)

Let B be a one-dimensional Brownian motion and $\Pi = \{0 = t_0 < \cdots < t_N = T\}$, with $\Delta t_k := t_{k+1} - t_k$, $\Delta_k B := B_{t_{k+1}} - B_{t_k}$ and mesh $|\Pi| := \max_k \Delta t_k$. Then

$$\sum_{k=0}^{N-1} (\Delta_k B)^2 \longrightarrow T \quad \text{in } L^2(\mathbb{P}) \text{ as } |\Pi| \to 0.$$

Importance Contrast: Deterministic squared increments $(\Delta t_k)^2$ clearly vanish. But $\Delta_k B \sim \mathcal{N}(0, \Delta t_k)$ has size $\sqrt{\Delta t_k}$, so over the $\approx T/\Delta t_k$ intervals:

$(\Delta_k B)^2 \sim \Delta t_k$	sums to $O(1)$	$\Rightarrow \ \mathrm{d}B_t \, \mathrm{d}B_t = \mathrm{d}t$
$\Delta_k B \Delta t_k \sim (\Delta t_k)^{3/2}$	sums to $O(\lvert\Pi\rvert^{1/2}) \to 0$	$\Rightarrow \ \mathrm{d}B_t \, \mathrm{d}t = 0$
$(\Delta t_k)^2$	sums to $O(\lvert\Pi\rvert) \to 0$	$\Rightarrow \ \mathrm{d}t \, \mathrm{d}t = 0$

In n dimensions, $\mathrm{d}B_t^i \, \mathrm{d}B_t^j = \delta_{ij} \, \mathrm{d}t$. Taylor expansions must be kept to **second order** in $\mathrm{d}B$.

The filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$

Definitions

- A σ-**algebra** $\mathcal{F}$ on Ω: a collection of events containing Ω, closed under complements and countable unions. $\mathbb{P}$ is a probability measure on $\mathcal{F}$.
- A **filtration**: σ-algebras with $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for all $s \leq t$.
- X is **adapted** if X_t is $\mathcal{F}_t$-measurable for every t.
- B is an $(\mathcal{F}_t)$-**Brownian motion** if adapted Brownian motion and $B_t - B_s$ is independent of $\mathcal{F}_s$.

How to read it

- $\mathcal{F}$: the questions about the experiment that we can assign a probability to.
- $\mathcal{F}_t$: the **information available at time** t. "Increasing" means we never forget.
- Canonical example: $\mathcal{F}_t = \sigma(B_s : s \leq t)$, everything decidable from the path up to time t.
- Adapted: "X_t is known at time t".

Technical: we assume the "usual conditions" ($\mathcal{F}_0$ contains the $\mathbb{P}$-null sets, $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$).

Terminology: martingales and semimartingales

Definitions

- An adapted M is a **martingale** if $\mathbb{E}|M_t| < \infty$ and $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ a.s., for all $s \leq t$.
- M is a **local martingale** if there are stopping times[1] $\tau_k \uparrow \infty$ such that each stopped process $(M_{t \wedge \tau_k})_{t \geq 0}$ is a martingale.
- X is a **continuous semimartingale** if $X_t = X_0 + M_t + A_t$, with M a continuous local martingale and A continuous, adapted, of finite variation, and $M_0 = A_0 = 0$.

Intuition

- **Martingale = "we expect to stay the same"**: given today's information, the expected future value is today's value. Examples: B_t and $B_t^2 - t$.
- **Local martingale**: a martingale on "local" finite horizons. It need not be a martingale.
- **Semimartingale = Martingale + Drift**: drift has finite variation so easy to deal with.

[1] τ is a stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ for all t: "whether we have stopped is known at time t".

Adapted, progressively measurable, predictable

Definitions (for a process $X : [0, \infty) \times \Omega \to \mathbb{R}^d$)

- **Adapted**: X_t is $\mathcal{F}_t$-measurable for every t.
- **Progressively measurable**: for every t, $(s, \omega) \mapsto X_s(\omega)$ on $[0, t] \times \Omega$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.
- **Predictable**: measurable for the σ-algebra generated by the left-continuous adapted processes.

Predictable $\Rightarrow$ progressively measurable $\Rightarrow$ adapted. Continuous and adapted $\Rightarrow$ predictable.

An easier way to read ($\mathcal{F}_t^B$ + light regularity)

- **Adapted**: $X_t = \varphi_t\big((B_s)_{s \in [0,t]}\big)$, a function of the path up to and including t.
- **Predictable**: $X_t = \varphi_t\big((B_s)_{s \in [0,t)}\big)$, a function of the path strictly before t.
- **Progressively measurable**: $X_t = \varphi_t\big(t, (B_s)_{s \in [0,t]}\big)$, a joint function of time and path.

Caveat: for continuous B, $(B_s)_{s<t}$ already determines B_t; the difference between $[0, t)$ and $[0, t]$ only bites for processes with jumps.

The Itô integral: elementary processes

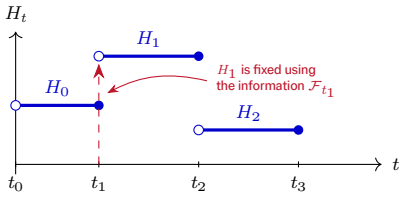
Definition (elementary process, its integral)

H is **elementary** if, for a partition Π of $[0,\, T]$,

$$H_t = \sum_{k=0}^{N-1} H_k \mathbf{1}_{(t_k, t_{k+1}]}(t),$$

with each H_k $\mathcal{F}_{t_k}$-**measurable** and in $L^2(\mathbb{P})$. Then

$$\int_0^T H_t \,\mathrm{d}B_t := \sum_{k=0}^{N-1} H_k \,(B_{t_{k+1}} - B_{t_k}) = \sum_{k=0}^{N-1} H_k \,\Delta_k B.$$



Constant on each $(t_k,\, t_{k+1}]$; the value is set at the **left endpoint** t_k.

Important Points

- H is **piecewise constant**: chosen with known information at the start of each interval.
- H is LEFT-continuous and adapted $\implies$ **predictable**.

The Itô integral: why the left endpoint?

Proposition (one-dimensional B; limits in $L^2(\mathbb{P})$ as $|\Pi| \to 0$)

$$\begin{aligned} \text{left endpoint (Itô):} \quad & \textstyle\sum_k B_{t_k}\,\Delta_k B \;\longrightarrow\; \tfrac{1}{2}\big(B_T^2 - T\big), \\ \text{right endpoint:} \quad & \textstyle\sum_k B_{t_{k+1}}\,\Delta_k B \;\longrightarrow\; \tfrac{1}{2}\big(B_T^2 + T\big), \\ \text{average (Stratonovich):} \quad & \textstyle\sum_k \tfrac{1}{2}\big(B_{t_k} + B_{t_{k+1}}\big)\,\Delta_k B \;=\; \tfrac{1}{2}B_T^2 \quad . \end{aligned}$$

Proof: Appendix A3.

Takeaways

- Left and right sums differ by $\sum_k (\Delta_k B)^2 \to T$: **the evaluation point changes the limit**. This cannot happen for smooth integrators.
- **Left endpoint**: H_k is independent of $\Delta_k B$, so the integral has **mean zero** and is a **martingale**.
- **Itô** = left endpoint: a martingale, with an extra term in the chain rule. **Stratonovich** = midpoint average: recovers ordinary calculus rules

The Itô integral: the most general integrand

Extension in three steps

Construction of $\int_0^t H_s \, \mathrm{d}B_s$

1. **Elementary** H: defined directly. It satisfies the Itô isometry $\mathbb{E}\left| \int_0^T H \, \mathrm{d}B \right|^2 = \mathbb{E} \int_0^T |H_t|^2 \, \mathrm{d}t$.
2. L^2 **integrands**: H progressively measurable with $\mathbb{E} \int_0^T |H_t|^2 \, \mathrm{d}t < \infty$. Elementary $H^{(m)} \to H$ exist; by the isometry, $\int H^{(m)} \mathrm{d}B$ is Cauchy in $L^2(\mathbb{P})$; define $\int H \mathrm{d}B$ as its limit. $\Rightarrow$ a **continuous square-integrable martingale**.
3. **Localisation**: H progressively measurable with $\int_0^T |H_t|^2 \, \mathrm{d}t < \infty$ a.s. Apply step 2 up to $\tau_k := \inf\{t : \int_0^t |H_s|^2 \mathrm{d}s \geq k\} \wedge T$ and let $k \to \infty$. $\Rightarrow$ a **continuous local martingale**.

The Story Each step **weakens the condition on** H, and allows for more general integrands (elementary $\to$ L^2). This weakens what we can say about the integral. (martingale $\to$ local martingale).

The Itô integral: the most general integrator

Integrating against continuous semimartingales

Definition

For $X = X_0 + M + A$ and H progressively measurable with $\int_0^T |H_t|^2 \,\mathrm{d}\langle M, M\rangle_t + \int_0^T |H_t|\,|\mathrm{d}A_t| < \infty$ a.s.,

$$\int_0^t H_s \,\mathrm{d}X_s := \int_0^t H_s \,\mathrm{d}M_s + \int_0^t H_s \,\mathrm{d}A_s.$$

Itô processes (all we need): $X_t = X_0 + \int_0^t b_s \,\mathrm{d}s + \int_0^t \sigma_s \,\mathrm{d}B_s$, with b, σ progressively measurable and $\int_0^T \big(|b_t| + |\sigma_t|^2\big)\mathrm{d}t < \infty$ a.s.: $\int_0^t H_s \,\mathrm{d}X_s = \int_0^t H_s b_s \,\mathrm{d}s + \int_0^t H_s \sigma_s \,\mathrm{d}B_s$.

The important takeaway Easy to integrate separately against martingale M and finite variation A.

Itô SDE solutions are semimartingales, so we need to integrate against these.

Zero mean

Proposition (zero mean)

If H is progressively measurable with $\mathbb{E}\int_0^T |H_t|^2\,\mathrm{d}t < \infty$, then $\mathbb{E}\big[\int_0^T H_t\,\mathrm{d}B_t\big] = 0$. More strongly, $t \mapsto \int_0^t H_s\,\mathrm{d}B_s$ is a martingale.

Proof intuition (elementary $\to$ limit) $\mathbb{E}\big[H_k\,\Delta_k B\big] = \mathbb{E}\big[H_k\,\mathbb{E}[\Delta_k B \mid \mathcal{F}_{t_k}]\big] = 0$, because H_k is $\mathcal{F}_{t_k}$-measurable (**the left endpoint**) and $\Delta_k B$ is independent of $\mathcal{F}_{t_k}$. For general H, L^2 convergence passes the zero mean to the limit.

Why it is useful Taking expectations **removes the noise term**. For $\mathrm{d}X_t = b_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t$:

$$\mathbb{E}X_t = \mathbb{E}X_0 + \int_0^t \mathbb{E}\,b_s\,\mathrm{d}s,$$

a deterministic equation. This is a key step in Dynkin's formula (Part II).

The Itô isometry

Proposition (Itô isometry)

If H is progressively measurable, $\mathbb{R}^{m\times n}$-valued, with $\mathbb{E}\int_0^T |H_t|^2\,\mathrm{d}t < \infty$ ($|\cdot|$ the Frobenius norm), then

$$\mathbb{E}\left|\int_0^T H_t\,\mathrm{d}B_t\right|^2 = \int_0^T \mathbb{E}|H_t|^2\,\mathrm{d}t.$$

Proof intuition ($m = n = 1$) Expand $\left(\sum_k H_k \Delta_k B\right)^2$ and condition on $\mathcal{F}_{t_k}$. Diagonal terms: $\mathbb{E}\left[H_k^2(\Delta_k B)^2\right] = \mathbb{E}[H_k^2]\,\Delta t_k$. Cross terms ($j < k$): $\mathbb{E}\left[H_j \Delta_j B\, H_k\, \mathbb{E}[\Delta_k B \mid \mathcal{F}_{t_k}]\right] = 0$. Sum: $\sum_k \mathbb{E}[H_k^2]\,\Delta t_k = \int_0^T \mathbb{E}H_t^2\,\mathrm{d}t$.

Why it is useful

1. It converts the second moment of a **random integral** into an ordinary integral of the **deterministic** quantity $\mathbb{E}|H_t|^2$. E.g. for deterministic h: $\mathrm{Var}\left(\int_0^t h(s)\,\mathrm{d}B_s\right) = \int_0^t h(s)^2\,\mathrm{d}s$.
2. It drives the construction: $H \mapsto \int H\,\mathrm{d}B$ and helps the generalisation of the Itô Integral.

Stochastic differential equations

Theorem (existence and uniqueness of strong solutions)

Let ξ be $\mathcal{F}_0$-measurable. Suppose $\exists K < \infty$ such that $\forall t \in [0,\, T]$, $x, y \in \mathbb{R}^n$:

$$\text{(Lip)}\ \ |b_t(x) - b_t(y)| + |\sigma_t(x) - \sigma_t(y)| \le K|x - y|, \qquad \text{(LG)}\ \ |b_t(x)| + |\sigma_t(x)| \le K(1 + |x|).$$

Then there is a unique **strong solution** of

$$X_t = \xi + \int_0^t b_s(X_s)\,\mathrm{d}s + \int_0^t \sigma_s(X_s)\,\mathrm{d}B_s$$

that is continuous and adapted. If $\mathbb{E}|\xi|^2 < \infty$, then $\mathbb{E}\sup_{t \le T}|X_t|^2 < \infty$.

How to read it

- **(Lip)** $\Rightarrow$ **uniqueness**: small differences cannot be amplified too fast, so solutions cannot separate.
- **(LG)** $\Rightarrow$ **no explosion**: coefficients grow at most linearly, so solutions cannot reach infinity in finite time.

Itô's formula: the chain rule for functions $f(t, X_t)$ of the processes so far

Theorem (Itô's formula; proof sketch in A4)

Let $X = (X^1, \dots, X^n)$ be a continuous semimartingale and $f \in C^{1,2}([0,\infty) \times \mathbb{R}^n; \mathbb{R})$. Then a.s., for all $t \geq 0$,

$$f(t, X_t) = f(0, X_0) + \int_0^t \partial_s f(s, X_s)\, \mathrm{d}s + \sum_i \int_0^t \partial_i f(s, X_s)\, \mathrm{d}X_s^i + \frac{1}{2} \sum_{i,j} \int_0^t \partial_i \partial_j f(s, X_s)\, \mathrm{d}\langle X^i, X^j \rangle_s.$$

For an Itô process $\mathrm{d}X_t = b_t\, \mathrm{d}t + \sigma_t\, \mathrm{d}B_t$, with $\Sigma_t := \sigma_t \sigma_t^\top$,

$$\mathrm{d}f(t, X_t) = \Big(\partial_t f + \langle b_t, \nabla f \rangle + \tfrac{1}{2} \operatorname{tr}\left(\Sigma_t \nabla^2 f\right) \Big)(t, X_t)\, \mathrm{d}t + \nabla f(t, X_t)^\top \sigma_t\, \mathrm{d}B_t.$$

How to read it

- The first three terms are the ordinary chain rule. The last is new: $(\mathrm{d}X)^2 \neq 0$, as $\mathrm{d}\langle X^i, X^j \rangle_t = (\Sigma_t)_{ij}\, \mathrm{d}t$.
- **Importance of** $C^{1,2}$: the most useful tool $\Rightarrow$ need to confirm regularity.

Further facts (I)

Quadratic covariation and the Itô product rule

Quadratic covariation

For continuous local martingales M, N there is a unique continuous adapted finite-variation process $\langle M, N\rangle$, $\langle M, N\rangle_0 = 0$, such that $MN - \langle M, N\rangle$ is a local martingale. For semimartingales X, Y, set $\langle X, Y\rangle := \langle M^X, M^Y\rangle$ (drifts do not contribute). Then $\sum_k \Delta_k X \Delta_k Y \to \langle X, Y\rangle_t$ in probability.

E.g. $\langle B^i, B^j\rangle_t = \delta_{ij} t$; for Itô processes $\langle X^i, X^j\rangle_t = \int_0^t (\Sigma_s)_{ij}\,\mathrm{d}s$.

Itô product rule

For continuous semimartingales X, Y: $\mathrm{d}(X_t Y_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}\langle X, Y\rangle_t$.

Intuition

- $\langle X, Y\rangle$ is the rigorous form of the rules $\mathrm{d}B_t\,\mathrm{d}B_t = \mathrm{d}t$, $\mathrm{d}B_t\,\mathrm{d}t = 0$.
- The product rule is Itô's formula for $f(x, y) = xy$. If X or Y has finite variation, $\langle X, Y\rangle = 0$: the ordinary product rule.

Further facts (II)

Lévy's characterisation and deterministic time change

Theorem (Lévy's characterisation)

A continuous local martingale M in $\mathbb{R}^n$ with $M_0 = 0$ and $\langle M^i, M^j \rangle_t = \delta_{ij}\, t$ is an $(\mathcal{F}_t)$-Brownian motion.

Why it is useful To show a process is a Brownian motion, check only the **martingale property and the quadratic variation**, not the law of its increments. Used to show that the shifted process in Girsanov's theorem is a Brownian motion.

Proposition (deterministic time change)

If $s \mapsto t(s)$ is a continuous increasing bijection of $[0, \infty)$ and M a continuous local martingale, then $s \mapsto M_{t(s)}$ is a continuous local martingale for $(\mathcal{F}_{t(s)})_{s \geq 0}$, with quadratic variation $\langle M, M \rangle_{t(s)}$.

Why it is useful It **changes the clock**, but **does not change the properties.**

Girsanov's theorem

Theorem (Girsanov; Gaussian intuition in A5)

Let B be an n-dimensional $(\mathcal{F}_t)$-Brownian motion under $\mathbb{P}$ and θ an $\mathbb{R}^n$-valued progressively measurable process with $\mathbb{E}_{\mathbb{P}} \exp\left(\frac{1}{2}\int_0^T |\theta_t|^2 \mathrm{d}t\right) < \infty$ (Novikov). Then the stochastic exponential $\mathcal{E}_t := \exp\left(\int_0^t \theta_s^\top \, \mathrm{d}B_s - \frac{1}{2}\int_0^t |\theta_s|^2 \, \mathrm{d}s\right)$, $t \in [0, T]$, is a $\mathbb{P}$-martingale with $\mathbb{E}_{\mathbb{P}}\, \mathcal{E}_T = 1$, and

(1) New measure: $\mathrm{d}\mathbb{Q} := \mathcal{E}_T \, \mathrm{d}\mathbb{P}$ is a probability measure on $\mathcal{F}_T$, equivalent to $\mathbb{P}$; $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}\,|_{\mathcal{F}_t} = \mathcal{E}_t$.

(2) New Brownian motion: $\tilde{B}_t := B_t - \int_0^t \theta_s \, \mathrm{d}s$ is an $(\mathcal{F}_t)$-Brownian motion on $[0, T]$ under $\mathbb{Q}$.

How to read Girsanov (1) is the density that **changes measure**. (2): under $\mathbb{Q}$, $B = \tilde{B} + \int_0^\cdot \theta_s \, \mathrm{d}s$ is a Brownian motion **with drift** θ: the measure changes the drift, never the noise.

Part II

First further theory: Markov semigroups, generators, invariant measures

Markov processes and semigroups

Time-homogeneous case: b and σ do not depend on t

Definitions

- X is **Markov** if $\mathbb{E}[\varphi(X_t) \mid \mathcal{F}_s] = \mathbb{E}[\varphi(X_t) \mid X_s]$ for all $s \le t$ and bounded measurable φ.
- Its **transition density** $p_t(x, y)$ satisfies $\mathbb{P}(X_{s+t} \in A \mid X_s = x) = \int_A p_t(x, y)\,\mathrm{d}y$.
- Its **Markov semigroup** is $P_t\varphi(x) := \mathbb{E}[\varphi(X_t) \mid X_0 = x] = \int \varphi(y)\, p_t(x, y)\,\mathrm{d}y$. It is linear and positive, with $P_0 = \mathrm{Id}$, $P_t 1 = 1$, and $P_{t+s} = P_t P_s$ (Chapman–Kolmogorov).

How to read these

- Markov: **given the present, the past adds no information about the future**. Solutions of SDEs are Markov.
- $P_t\varphi(x)$: the **expected value of φ after time t, started from x**. The Markov property reads $\mathbb{E}[\varphi(X_t) \mid \mathcal{F}_s] = P_{t-s}\varphi(X_s)$.
- On laws: if $X_0 \sim \nu$, then $X_t \sim \nu P_t$, where $\int \varphi\,\mathrm{d}(\nu P_t) := \int P_t\varphi\,\mathrm{d}\nu$.

The generator and Dynkin's formula

Definition and proposition

The **generator** is $\mathcal{L}\varphi(x) := \lim_{h\downarrow 0} \frac{1}{h}\big(P_h\varphi(x) - \varphi(x)\big)$, where the limit exists. For $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}B_t$ well posed, $\Sigma := \sigma\sigma^\top$, and $\varphi \in C_c^2(\mathbb{R}^n)$:

$$\mathcal{L}\varphi = \langle b, \nabla\varphi\rangle + \tfrac{1}{2}\operatorname{tr}\big(\Sigma\nabla^2\varphi\big), \qquad \mathbb{E}\varphi(X_t) - \mathbb{E}\varphi(X_0) = \int_0^t \mathbb{E}\big[\mathcal{L}\varphi(X_s)\big]\,\mathrm{d}s \quad \text{(Dynkin)}.$$

Proof idea: Itô's formula with $f = \varphi$ gives $\varphi(X_t) - \varphi(X_0) = \int_0^t \mathcal{L}\varphi(X_s)\,\mathrm{d}s + \int_0^t \nabla\varphi(X_s)^\top \sigma(X_s)\,\mathrm{d}B_s$, and the stochastic integral is of zero mean.

Why it is useful A **single deterministic object** $\mathcal{L}$ governs the evolution of expectations, $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\varphi(X_t) = \mathbb{E}\,\mathcal{L}\varphi(X_t)$. This can read $\partial_t P_t\varphi = P_t\mathcal{L}\varphi$. With more regularity: $\partial_t P_t\varphi = \mathcal{L}P_t\varphi$ which is Kolmogorov's backward equation.

In this module: the Langevin diffusion has generator $\mathcal{L}\varphi = \Delta\varphi - \langle\nabla U, \nabla\varphi\rangle$.

Invariant measures

Definition and relation to the generator

A probability measure μ is **invariant** if $\mu P_t = \mu$ for all $t \geq 0$, i.e. $X_0 \sim \mu \Rightarrow X_t \sim \mu$ for all t.

If μ is invariant, then $\int \mathcal{L}\varphi \, \mathrm{d}\mu = 0$ for all $\varphi \in C_c^\infty(\mathbb{R}^n)$, written $\mathcal{L}^*\mu = 0$.

Proof: Dynkin with $X_0 \sim \mu$ gives $0 = \mathbb{E}\varphi(X_t) - \mathbb{E}\varphi(X_0) = \int_0^t \mathbb{E}\, \mathcal{L}\varphi(X_s) \, \mathrm{d}s = t \int \mathcal{L}\varphi \, \mathrm{d}\mu$.

The converse ($\mathcal{L}^*\mu = 0 \Rightarrow \mu$ invariant) can be proven with some extra regularity

Why it is useful

- Invariance turns **sampling into simulation**: to sample from μ, run dynamics that leave μ invariant for a long time.
- Checking invariance only needs $\mathcal{L}$.

Appendix

Statements for reference

A1. Divergences

Definition (Kullback–Leibler divergence)

For probability measures μ, ν:

$$\mathrm{KL}(\mu \,\|\, \nu) := \begin{cases} \displaystyle\int \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\, \mathrm{d}\mu & \text{if } \mu \ll \nu, \\ +\infty & \text{otherwise.} \end{cases}$$

$\mathrm{KL}(\mu\|\nu) \geq 0$, with equality if and only if $\mu = \nu$ (Jensen). It is **not symmetric** and not a metric.

How to read it The expected log-likelihood ratio under μ: how distinguishable μ is from ν when the data come from μ. Maximum likelihood is KL minimisation.

Further results: total variation, χ^2 e.g. see Y. Polyanskiy, Y. Wu, Information Theory: From Coding to Learning

A2. The Fokker–Planck equation

Proposition (Fokker–Planck equation)

Let X solve $\mathrm{d}X_t = b_t(X_t)\,\mathrm{d}t + \sigma_t(X_t)\,\mathrm{d}B_t$, $\Sigma_t := \sigma_t\sigma_t^\top$, and suppose X_t has a smooth density ρ_t. Then

$$\partial_t \rho_t = -\nabla\cdot(b_t\rho_t) + \frac{1}{2}\sum_{i,j=1}^{n} \partial_i\partial_j\big((\Sigma_t)_{ij}\,\rho_t\big) =: \mathcal{L}_t^*\rho_t.$$

For additive noise $\sigma_t = \sqrt{2}\,I_n$: $\partial_t\rho_t = \nabla\cdot\big(\rho_t(\nabla\log\rho_t - b_t)\big)$.

Important takeaways

- The **dual of Dynkin**: $\mathcal{L}$ acts on test functions (expectations), $\mathcal{L}^*$ acts on densities (laws). Derivation: $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\,\rho_t = \int(\mathcal{L}_t\varphi)\,\rho_t$, then integrate by parts.
- Langevin, $b = -\nabla U$: for $\rho \propto e^{-U}$ the flux $\rho(\nabla\log\rho - b)$ vanishes, so ρ is stationary.

Further results: see e.g. G. A. Pavliotis, Stochastic Processes and Applications (Springer, 2014), Ch. 4.

A3. Proofs: quadratic variation and left-endpoint sums

Quadratic variation in L^2

Let $Q_\Pi := \sum_k (\Delta_k B)^2$. Then $\mathbb{E}\, Q_\Pi = \sum_k \Delta t_k = T$. The increments are independent and $\mathrm{Var}\left((\Delta_k B)^2\right) = 2(\Delta t_k)^2$ (since $\mathbb{E} Z^4 = 3\sigma^4$ for $Z \sim \mathcal{N}(0, \sigma^2)$), so

$$\mathbb{E}\,|Q_\Pi - T|^2 = \mathrm{Var}(Q_\Pi) = 2\sum_k (\Delta t_k)^2 \le 2\,T\,|\Pi| \longrightarrow 0.$$

Left, right and average sums

Telescoping, $B_T^2 = \sum_k \left(B_{t_{k+1}}^2 - B_{t_k}^2\right) = \sum_k \left(2B_{t_k}\Delta_k B + (\Delta_k B)^2\right)$. Hence

$$\sum_k B_{t_k}\Delta_k B = \tfrac{1}{2}B_T^2 - \tfrac{1}{2}Q_\Pi \xrightarrow{L^2} \tfrac{1}{2}\left(B_T^2 - T\right), \qquad \sum_k B_{t_{k+1}}\Delta_k B = \sum_k B_{t_k}\Delta_k B + Q_\Pi \xrightarrow{L^2} \tfrac{1}{2}\left(B_T^2 + T\right),$$

and $\sum_k \tfrac{1}{2}\left(B_{t_k} + B_{t_{k+1}}\right)\Delta_k B = \tfrac{1}{2}\sum_k \left(B_{t_{k+1}}^2 - B_{t_k}^2\right) = \tfrac{1}{2}B_T^2$. So $\int_0^T B_t \,\mathrm{d}B_t = \tfrac{1}{2}(B_T^2 - T)$, of mean zero.

A4. Itô's formula: proof sketch

One dimension, $f = f(x) \in C^2$, $\mathrm{d}X_t = b_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t$

Partition $[0, t]$, set $\Delta_k X := X_{t_{k+1}} - X_{t_k}$, telescope and Taylor expand to second order:

$$f(X_t) - f(X_0) = \underbrace{\sum_k f'(X_{t_k})\,\Delta_k X}_{(1)} + \underbrace{\frac{1}{2}\sum_k f''(X_{t_k})\,(\Delta_k X)^2}_{(2)} + \underbrace{\sum_k R_k}_{(3)}, \quad |R_k| \le \omega\big(|\Delta_k X|\big)(\Delta_k X)^2,$$

with ω a modulus of continuity of f'' on the (compact) range of the path. As $|\Pi| \to 0$, in probability:

The three limits

1. $(1) \to \int_0^t f'(X_s)\,\mathrm{d}X_s$: these are **left-endpoint sums**, i.e. the Itô integral.
2. $(2) \to \frac{1}{2}\int_0^t f''(X_s)\,\sigma_s^2\,\mathrm{d}s$: $(\Delta_k X)^2 \approx \sigma_{t_k}^2 (\Delta_k B)^2 \approx \sigma_{t_k}^2 \Delta t_k$ by **quadratic variation**; drift parts are $O\big((\Delta t_k)^{3/2}\big)$.
3. $(3) \to 0$: $|(3)| \le \max_k \omega(|\Delta_k X|) \sum_k (\Delta_k X)^2$; the max $\to 0$ by **path continuity**, the sum stays bounded.

Hence $f(X_t) = f(X_0) + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\,\sigma_s^2\,\mathrm{d}s$. (Time dependence adds $\int_0^t \partial_s f\,\mathrm{d}s$.)

A5. Girsanov's theorem: intuition

Changing the measure changes the drift

Key point: "B is a Brownian motion" is a statement about B under a particular measure $\mathbb{P}$. Under another measure, the same process can have a drift.

Toy example: one Gaussian

Let $Z \sim \mathcal{N}(0,1)$ under $\mathbb{P}$, $\theta \in \mathbb{R}$, and define $\mathbb{Q}$ by $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} := e^{\theta Z - \theta^2/2}$. For bounded φ,

$$\mathbb{E}_{\mathbb{Q}}[\varphi(Z)] = \int \varphi(z)\, e^{\theta z - \theta^2/2}\, \frac{e^{-z^2/2}}{\sqrt{2\pi}}\, \mathrm{d}z = \int \varphi(z)\, \frac{e^{-(z-\theta)^2/2}}{\sqrt{2\pi}}\, \mathrm{d}z,$$

so under $\mathbb{Q}$, $Z \sim \mathcal{N}(\theta, 1)$: equivalently $Z - \theta \sim \mathcal{N}(0,1)$.

The motivating takeaway Reweighting by an **exponential density** shifts the mean; Z itself is unchanged. For paths, replace Z by the path of B and θ by a drift θ_t: under the reweighted measure, B is a Brownian motion with drift θ, i.e. $B - \int_0^{\cdot} \theta_s \, \mathrm{d}s$ is a Brownian motion.

Further reading

- J.-F. Le Gall, Brownian Motion, Martingales, and Stochastic Calculus, Springer, 2016.
- B. Øksendal, Stochastic Differential Equations, 6th ed., Springer, 2003.
- I. Karatzas, S. E. Shreve, Brownian Motion and Stochastic Calculus, 2nd ed., Springer, 1991.
- D. Revuz, M. Yor, Continuous Martingales and Brownian Motion, 3rd ed., Springer, 1999.
- G. A. Pavliotis, Stochastic Processes and Applications, Springer, 2014.